

PROJECT COMPLETION REPORT

Project Title: BankBot: an AI Chatbot for Banking FAQs

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Internship Organization: Infosys Springboard Virtual Internship 6.0

Batch No. : 4

Submission Date: December 11, 2025

1. Executive Summary

BankBot is a comprehensive, "Neo-FinTech" banking prototype developed as the Capstone Project for the Infosys Springboard Virtual Internship. The system is engineered to modernize traditional transactional banking by seamlessly integrating it with advanced conversational Artificial Intelligence through a **Hybrid Intelligence System**.

This system distinguishes itself from conventional generative chatbots by integrating two fundamental components:

1. **Rule-Based Logic:** This component is essential for executing secure, deterministic financial operations, including, but not limited to, fund transfers and strict adherence to Know Your Customer (KYC) protocols.
2. **Machine Learning:** Utilizing TF-IDF and Logistic Regression, this functionality ensures flexible and highly accurate intent recognition, thereby facilitating a conversational and readily accessible user experience.

This strategic amalgamation guarantees the requisite safety and security for managing sensitive financial data while simultaneously sustaining a user-centric, conversational interface.

The application is characterized by a secure, role-based architecture and features a sophisticated, glassmorphic "Deep Void" user interface. It meticulously simulates a realistic banking environment, enabling users to manage their accounts, review comprehensive transaction histories, and access the continuous 24/7 AI assistant.

2. Project Scope & Architecture

The project was executed over an 8-week timeline, divided into four distinct technical milestones:

Milestone 1: NLU Engine Development

- **Objective:** Develop the "Brain" of the chatbot using Natural Language Understanding (NLU).
- **Outcome:** Implemented a Scikit-Learn pipeline using TF-IDF vectorization and Logistic Regression to classify user intents (e.g., `check_balance`, `transfer_money`, `lost_card`) with high confidence.

Milestone 2: Dialogue Management System

- **Objective:** Create a deterministic logic flow to handle complex queries.
- **Outcome:** Built a custom State Machine and "Smart Routing" system to manage conversation context. This ensures the bot remembers previous inputs (e.g., distinguishing between a menu selection "1" and a transfer amount "1").

Milestone 3: UI Integration (Neo-FinTech Frontend)

- **Objective:** Develop a responsive, modern web interface.
- **Outcome:** Created a Flask-based web server rendering HTML5/Bootstrap templates. The design utilizes a "Deep Void" (`#020617`) and "Electric Purple" (`#7c3aed`) theme with glassmorphism effects to provide a premium user experience.

Milestone 4: Admin Panel & Database Integration

- **Objective:** Connect the frontend to a persistent backend.
- **Outcome:** Integrated an SQLite database to handle User Data, Transaction Ledgers, and Chat Logs. Developed a Corporate Admin Portal for system monitoring and dynamic knowledge base updates.

Technology Stack

- **Frontend:** HTML5, CSS3 (Glassmorphism), JavaScript, Bootstrap, Dash.
- **Backend:** Python, Flask (Web Server).
- **AI/ML:** Scikit-Learn (TF-IDF, Logistic Regression), Custom State Machine.
- **Database:** SQLite.
- **Tools:** Git, GitHub, VS Code.

3. Implementation Details

3.1 The Hybrid Intelligence Engine

The core innovation of BankBot is its hybrid approach to dialogue management:

- **General Inquiries:** Handled by the ML model (Logistic Regression), allowing users to ask questions naturally (e.g., "How do I block my card?").
- **Sensitive Operations:** Handled by a deterministic State Machine. When a user initiates a transfer, the bot locks into a specific "state," ignoring general chit-chat until the transaction is confirmed or cancelled. This prevents "AI hallucinations" regarding financial data.

3.2 Secure Database Architecture

The application utilizes a relational SQLite database with three primary schemas:

1. **User Table:** Stores hashed credentials, account balances, and KYC status.
2. **Transaction Ledger:** Records every financial movement with precise timestamps and status flags (Success/Failed).
3. **Chat Logs:** Archives user-bot interactions for the Admin Nexus to facilitate model retraining.

3.3 Role-Based Access Control (RBAC)

The system implements strict session management to separate environments:

- **Retail Portal:** For standard customers (Sia, Suriya, Bhaskar) to perform banking tasks.
- **Corporate Portal:** For System Administrators (`admin_bot`) to view analytics and manage the AI training data.

4. System Walkthrough & Outputs

This section demonstrates the functional modules of the Bank of Trust platform.

4.1 Landing & Authentication

The system utilizes a responsive landing page, designed with the "Neo-FinTech" aesthetic, as its primary entry point. This page provides separate access routes tailored for both Personal Banking users and Corporate Administration.

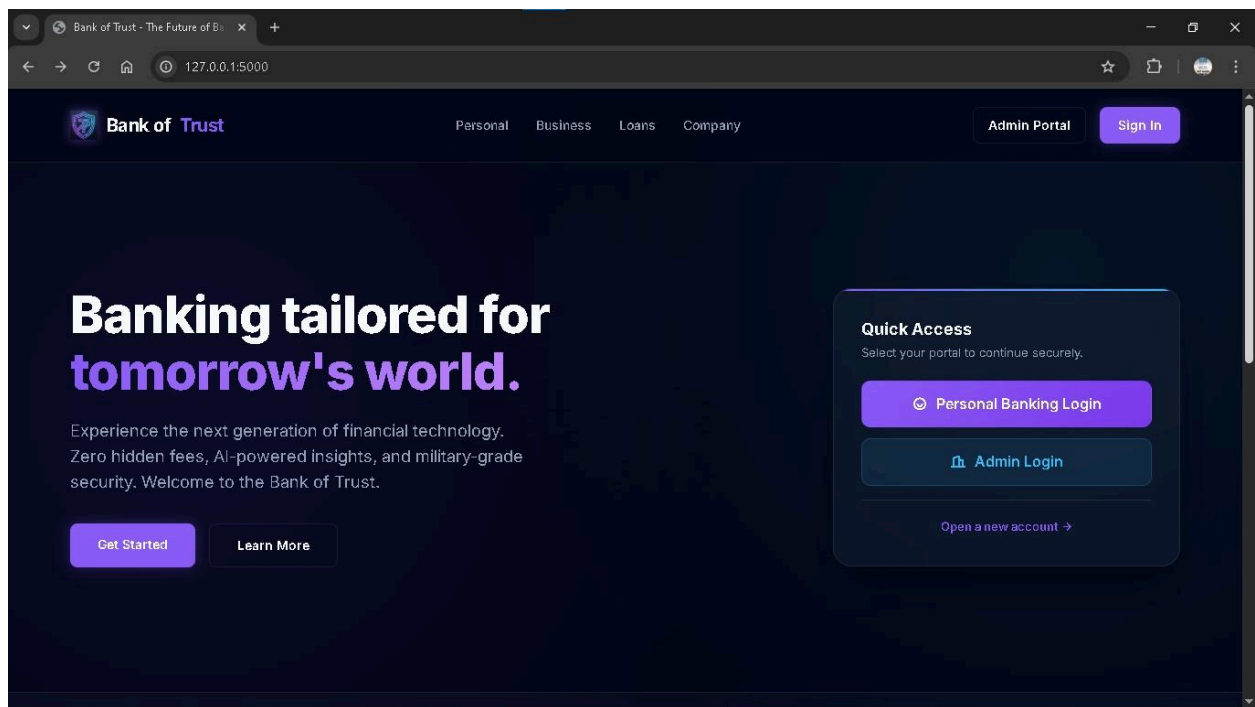


Figure 1: Home Page & Corporate Login (*The landing page serves as the gateway to the secure portals*)

4.2 Customer Dashboard & AI Interface

Upon successful login, the user is directed to the Dashboard. The AI Banking Assistant is seamlessly integrated into the interface, facilitating efficient multitasking; users are able to review their account balance through the graphical user interface or by querying the assistant.

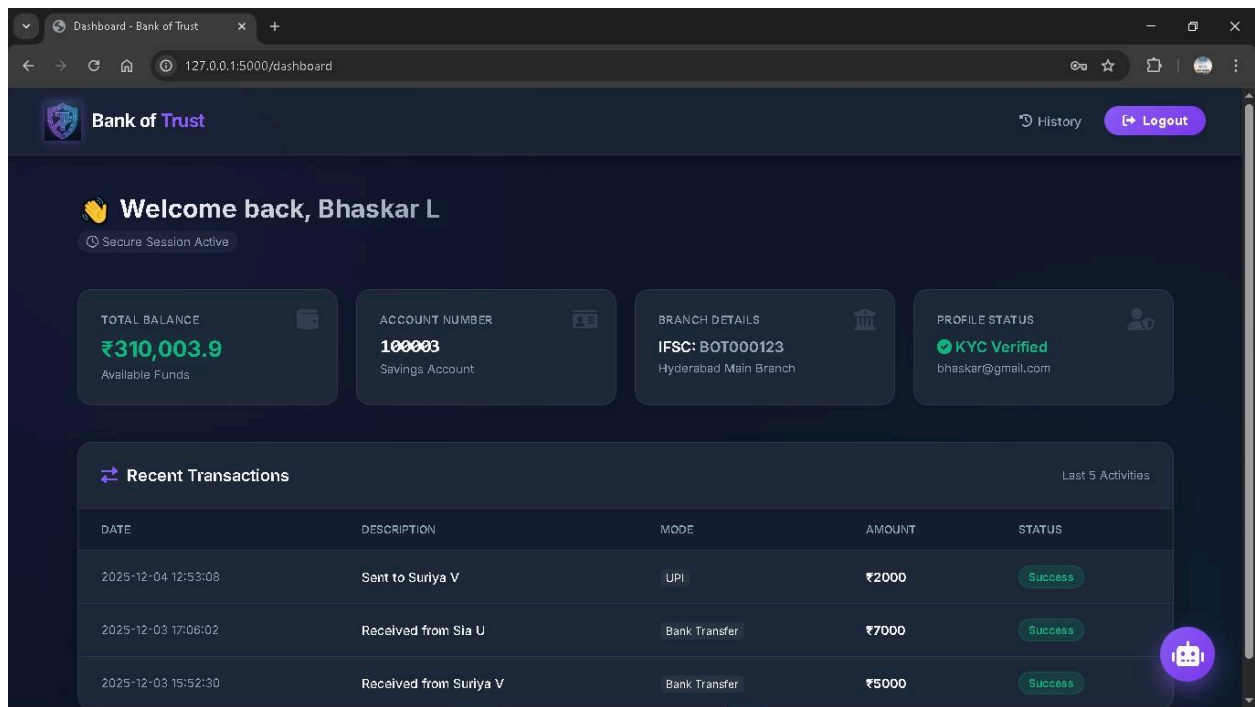


Figure 2: Customer Dashboard with Active AI Session (Displays real-time balance, recent transactions, and the floating Chatbot window)

4.3 Conversational Banking

The conversational interface possesses the capability to process natural language. As subsequently demonstrated, the system accurately discerns the user's intent and provides a contextually suitable response.

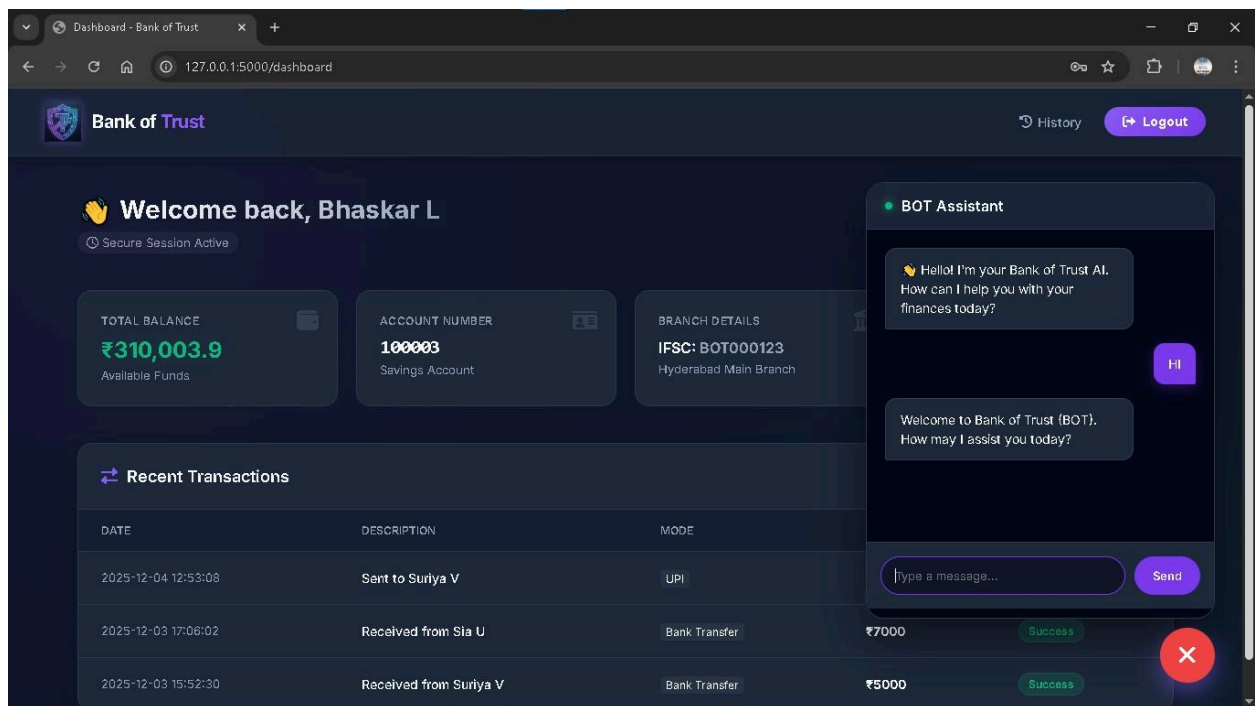


Figure 3: AI Chat Interface *(Demonstrating the bot's ability to handle greetings and specific intent queries)*

4.4 Data Persistence & Archives

A key feature of the backend is the **SQLite Session Manager**, which archives every interaction. This ensures that users can review previous support conversations and transactions even after logging out, proving the system's database integration is fully functional.

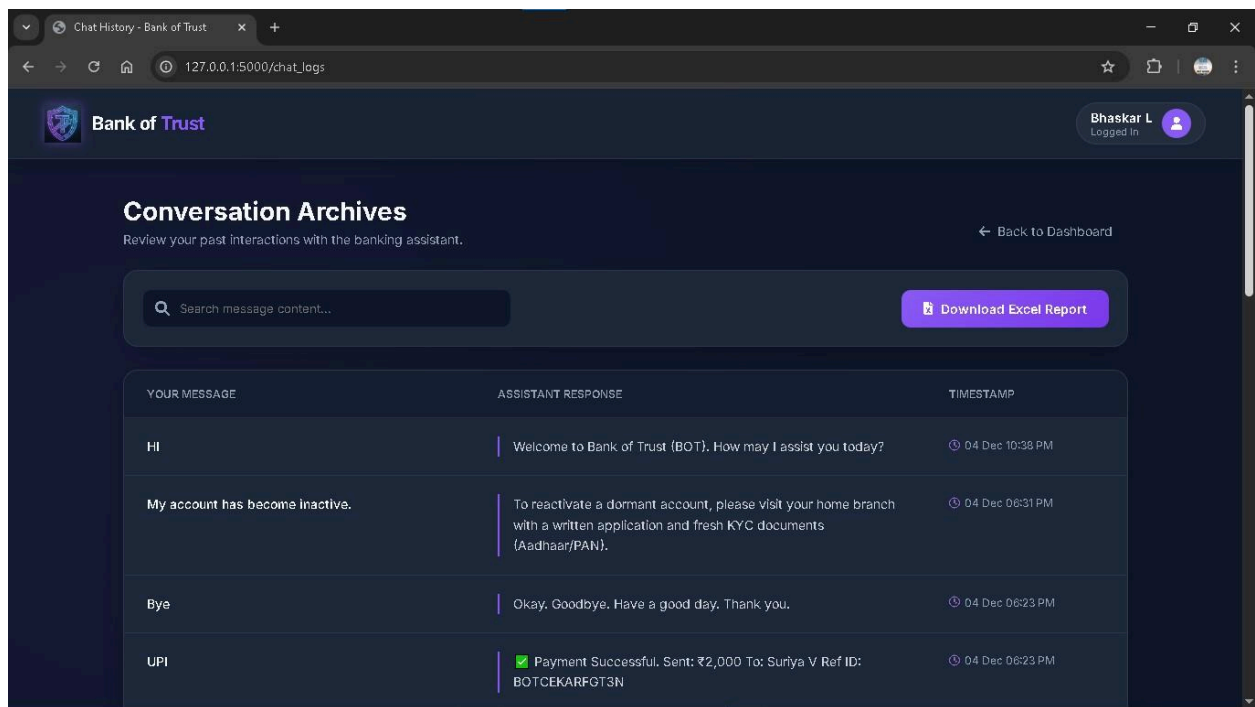


Figure 4: Customer Conversation Archives (The "History" tab retrieves past dialogue sessions and assistant responses from the database)

4.4 Administrative Control

The Admin Portal provides a high-level view of system health. It includes analytics on user intent distribution and confidence scores, ensuring the model's performance is monitored.

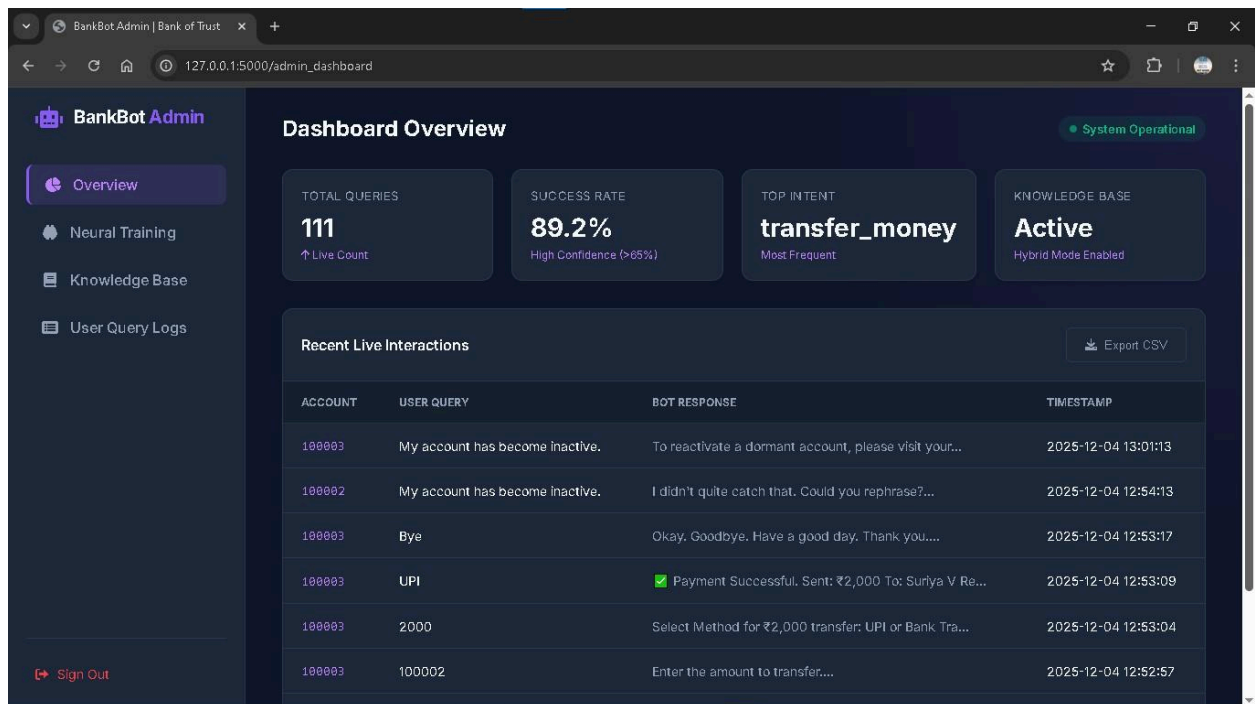


Figure 4: Admin Dashboard & Analytics (The "Model Training", "Knowledge Base" and "User Query Logs" sections allow for real-time system improvements)

4.6 Machine Learning Logs (The "Brain" View)

To ensure safety and accuracy, the system records the **Confidence Score** for all NLU predictions. The "User Query Logs" interface enables administrators to audit the classification performed by the Machine Learning model on specific inputs (e.g., matching "UPI" to transfer_money with 100% confidence).

BankBot Admin

Overview

Neural Training

Knowledge Base

User Query Logs

Sign Out

User Query Logs

System Operational

User Query Logs (Last 100)

Refresh

Export All

ID	TIME	ACCOUNT	USER MESSAGE	INTENT / CONFIDENCE	RESPONSE
#111	2025-12-04 13:01:13	100003	My account has become inactive.	faq_match 100%	To reactivate a dormant account, please visit your...
#110	2025-12-04 12:54:13	100002	My account has become inactive.	fallback 0%	I didn't quite catch that. Could you rephrase?...
#109	2025-12-04 12:53:17	100003	Bye	goodbye 57%	Okay, Goodbye. Have a good day. Thank you....
#108	2025-12-04 12:53:09	100003	UPI	transfer_money 100%	Payment Successful. Sent: ₹2,000 To: Suriya V Re...
#107	2025-12-04 12:53:04	100003	2000	transfer_money 100%	Select Method for ₹2,000 transfer: UPI or Bank Tra...
#106	2025-12-04 12:52:57	100003	100002	transfer_money 100%	Enter the amount to transfer....
#105	2025-12-04 12:52:48	100003	Transfer Money	transfer_money 100%	Invalid format. Please enter a valid 8-16 digit ac...
#104	2025-12-04 12:50:26	100003	100002	transfer_money 100%	Account 100002: Available

Figure 6: Real-Time User Query Logs (Detailed audit trail showing Input, Predicted Intent, Confidence Score, and Bot Response)

5. Challenges & Solutions

During the development lifecycle, several technical challenges were addressed:

1. Challenge: Intent Ambiguity (Smart Routing)

- *Issue:* The bot struggled to distinguish between a numerical menu choice (e.g., "Press 1 for Savings") and a numerical monetary amount (e.g., "Transfer 1 dollar").
- *Solution:* Implemented a **Context-Aware State Machine**. The bot checks the current active "State" before processing input. If the user is in the `TRANSFER_AMOUNT` state, all numerical inputs are treated as currency, overriding standard menu logic.

2. Challenge: Iframe Session Persistence

- *Issue:* Embedding the chatbot as an independent module within the dashboard caused session disconnects where the bot didn't know *who* was logged in.
- *Solution:* Utilized Flask's server-side session management to share the `user_id` context globally across both the main dashboard and the chat window iframe.

6. Code Repository

The complete source code, accompanying datasets, and detailed installation documentation are publicly accessible on GitHub.

- **Repository URL:** [https://github.com/\[author\]/BankBot_AI_Chatbot_for_Banking_FAQs](https://github.com/[author]/BankBot_AI_Chatbot_for_Banking_FAQs)

Project Components: This includes the core application files (app.py, dialogue_manager.py) and the bankbot_final_expanded1.csv training dataset.

7. Conclusion & Future Roadmap

The BankBot project conclusively demonstrates that a **Hybrid AI approach** represents the optimal solution for the FinTech sector, effectively balancing the adaptability inherent in Machine Learning with the enhanced security of rule-based systems. The current prototype is fully functional for local deployment and incorporates a comprehensive user and administrative workflow.

Future enhancements include:

- **WhatsApp Integration:** Extending the chatbot's reach via the Meta API.
- **Voice Banking:** Implementing STT/TTS (Speech-to-Text) for accessibility.
- **PDF Statement Generation:** allowing users to download official transaction records directly from the chat window.